GD&T Scheme and Tolerance Analysis of an Exoskeleton Motor Moving Side Plate

Datum Selection, Fixed-Fastener Allocation, Projected Zones and a Bracket Coplanarity Stack-Up

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Abstract

The Motor Moving Side Plate is a 12.7 mm (0.500 in) 6061-T6 plate that couples the rotor of a CubeMars AKH70-16 actuator to the next link of an upper-limb exoskeleton and carries two gusset brackets on its side edges. This report documents the ASME Y14.5-2018 drawing prepared for the part and the calculations behind each tolerance. Datum A is the face that seats on the rotor spacer and datum B is the six-hole bolt pattern that locates the plate on the rotor, referenced at maximum material condition (MMC). Every position tolerance is derived from the fixed-fastener relation $T_1 + T_2 = H - F$. The analysis shows that the plate's own tolerances are consistent, but it exposes one conflict: the already-machined ∅4.3 bracket holes leave the bracket only ∅0.1 of position tolerance. Opening them to ∅4.5 resolves it. A four-contributor stack-up predicts a worst-case offset of ±0.50 mm (RSS ±0.26 mm) between the two faces the bracket must bridge; aligning the parts with a straightedge before torquing removes the largest term.

1 Part function and what must be controlled

The side plate is the first part of the moving link (Figure 2). Its functional interfaces are:

(1) **Rotor interface.** The large face seats on a ∅50 rotating spacer and six M4 screws pass through ∅4.4 clearance holes on a ∅35 pitch circle into the rotor's threaded holes. A central ∅7.2 hole clears the rotor hub.

(2) **Link interface.** Four ∅4.4 holes near the lower edge take M4 screws into the end face of the Static Connector, a second 12.7 mm plate. The plate's lower edge sits flush with the Static Connector's outer face.

(3) **Bracket interface.** Two M4 threaded holes in each side edge hold an L-shaped gusset bracket (5 mm 6061 plate, ∅4.3 holes) whose other leg is bolted to the Static Connector. The companion FEA report shows that these brackets carry most of the joint torque, so they only work if they seat flat on both parts.

The tolerancing question is therefore: *how far can each feature move and still let every screw assemble and the bracket seat flat?* Sizes, locations and the outer profile are all set from that question rather than from generic title-block tolerances.

2 Datum reference frame

Datum A: the spacer-contact face. This face is the primary datum because it is the surface the part physically seats on in assembly; it removes three degrees of freedom (one translation, two rotations). It carries a flatness tolerance of $\boxed{\square \mid 0.1}$ so that the six screws clamp the plate without rocking. The plate is bought as 0.500 in stock and is not faced, so its thickness is dimensioned as `12.7 (0.500 STOCK)` rather than as a machined size.

Datum B: the 6×∅4.4 pattern at MMC. The secondary datum is the bolt pattern, not an outer edge, because the screws, not the edges, locate the part on the rotor. Choosing the edges would make inspection measure the part differently from the way it functions. The pattern is controlled as $\boxed{\oplus \mid \varnothing 0.2\,\text{Ⓜ} \mid \text{A}}$ and labelled datum B. Because B is a feature of size, it is referenced at MMC ($\text{B}\text{Ⓜ}$) in every downstream frame. The datum feature simulator is then a set of six fixed pins at the pattern's virtual condition,

$$\text{VC} = \text{MMC hole} - \text{position tolerance} = 4.40 - 0.20 = 4.20\,\text{mm}, \tag{1}$$

and any clearance between the actual holes and those pins is *datum shift*: extra freedom the inspector may use, exactly as the real screws allow the part to shift on the rotor. B removes the remaining two translations and the rotation about A, so no tertiary datum is needed.

3 Drawing summary

Table 1 lists every geometric control on the drawing. All locating dimensions for these features are basic (boxed); the title-block tolerances (one place ±0.2, two places ±0.1, angles ±0.5°) apply only to the remaining non-basic dimensions.

Table 1: Geometric controls on the Moving Side Plate drawing.

Feature	Control	Purpose
Spacer face (A)	$\boxed{\square \mid 0.1}$	Seats without rocking under six-screw clamp
$6\times\varnothing 4.4^{+0.15}_{0}$ (B)	$\boxed{\oplus \mid \varnothing 0.2\,\text{Ⓜ} \mid \text{A}}$	Assembles to rotor M4 holes; defines datum B
$\varnothing 7.2^{+0.3}_{0}$	$\boxed{\oplus \mid \varnothing 0.5 \mid \text{A} \mid \text{B}\text{Ⓜ}}$	Hub clearance only; loose tolerance
$4\times\varnothing 4.4^{+0.15}_{0}$	$\boxed{\oplus \mid \varnothing 0.2\,\text{Ⓜ} \mid \text{A} \mid \text{B}\text{Ⓜ}}$	Assembles to Static Connector M4 holes
4×M4×0.7-6H, 2 per side	$\boxed{\oplus \mid \varnothing 0.2\,\text{Ⓟ}5 \mid \text{A} \mid \text{B}\text{Ⓜ}}$	Bracket screws; zone projected through 5 mm bracket
Outer profile, all around	$\boxed{\frown \mid 0.2 \mid \text{A} \mid \text{B}\text{Ⓜ}}$	Side edges and lower edge seat the bracket and align with the Static Connector

Why profile instead of size tolerances on the outline. The outline is waterjet cut. A single all-around profile tolerance controls the location, orientation and form of the side edges, lower edge and $\boxed{\text{R35}}$ arc relative to the datum frame in one frame, which a set of ± dimensions cannot do: under Rule #1, a ± width says nothing about where the edges sit relative to the bolt pattern. A

zone of 0.2 (±0.1 mm) is near the edge of typical waterjet capability (±0.1 mm–±0.15 mm), which is why note 4 on the drawing requires the cut taper to stay inside the zone.

Symmetry. The 4×⌀4.4 pattern (18, 45) and the 70 width are dimensioned symmetrically about a vertical centerline drawn through the datum B pattern centre, which ties their horizontal location to datum B.

4 Fixed-fastener tolerance allocation

4.1 Relation

Every bolted interface on this part is a *fixed-fastener* case: the screw is threaded into one part (and therefore cannot move relative to it) and passes through a clearance hole in the other. For the screw to assemble when both parts are at MMC and their position errors are at the limit in opposite directions,

$$T_1 + T_2 = H - F, \tag{2}$$

where H is the MMC (smallest) clearance-hole diameter, F is the MMC (largest) fastener diameter (4.0 mm for M4), and T_1, T_2 are the diametral position tolerances of the clearance hole and the threaded hole. For comparison, the floating-fastener case (clearance holes in both parts, nut on the end) allows $T = H - F$ for *each* part. Fixed fasteners must share the budget, which is why they are the stricter case.

4.2 Allocation

Table 2: Fixed-fastener allocation for each M4 joint. All values in mm, diametral.

Joint	H	F	$H - F$	Side plate	Mating part
6× side plate → rotor threads	4.4	4.0	0.4	⌀0.2 Ⓜ	⌀0.2[a]
4× side plate → Static Connector	4.4	4.0	0.4	⌀0.2 Ⓜ	⌀0.2 Ⓟ[b]
Bracket (⌀4.3) → side-plate threads	4.3	4.0	0.3	⌀0.2 Ⓟ 5	**⌀0.1**
Bracket, reworked to ⌀4.5	4.5	4.0	0.5	⌀0.2 Ⓟ 5	⌀0.3 Ⓜ

[a] Actuator vendor tolerance, not published; assumed. Its projected height is the side plate plus spacer, $12.7 + 15 = 27.7$ mm.
[b] To be specified on the Static Connector drawing, projected 12.7 mm.

The two ⌀4.4 patterns split the 0.4 mm budget evenly. The bracket joint, however, has only 0.3 mm because the bracket holes were already machined at ⌀4.3 (a close-fit M4 clearance). With ⌀0.2 assigned to the side-plate threads, the bracket is left with ⌀0.1, which is not realistic for a hand-drilled or waterjet hole. Two fixes were considered:

- **Split 0.15 / 0.15.** Keeps the existing bracket, but tightens the side-plate threads, which are drilled into a 12.7 mm edge after the profile cut and are the hardest features on the part to hold.
- **Open the bracket holes to ⌀4.5 (adopted).** Enlarging an existing hole with a drill is a few minutes of rework, raises the budget to 0.5 mm, and gives the bracket ⌀0.3 at MMC while leaving the side plate unchanged.

4.3 Bonus tolerance

Because the clearance holes are toleranced at MMC, the allowed position error grows as the hole departs from MMC, while the virtual condition (4.20 mm) stays fixed (Table 3). A functional gauge with ⌀4.20 pins therefore accepts exactly the parts that assemble.

Table 3: Bonus tolerance for the $\varnothing 4.4^{+0.15}_{0}$ holes.

Actual hole size	Bonus	Allowed position	Virtual condition
4.40 (MMC)	0.00	⌀0.20	4.20
4.45	0.05	⌀0.25	4.20
4.50	0.10	⌀0.30	4.20
4.55 (LMC)	0.15	⌀0.35	4.20

5 Projected tolerance zone for the bracket threads

A position tolerance on a threaded hole normally applies over the depth of the thread. The screw, however, has to pass through the bracket *above* the hole, and a tilted thread axis keeps diverging outside the part (Figure 1).

With a ⌀0.2 zone over the 8 mm full-thread depth, the worst tilt runs from one side of the zone at the bottom of the thread to the other side at the surface, a slope of $0.2/8 = 0.025$. At height h above the surface the axis is offset from true position by

$$e(h) = 0.1 + 0.025\,h \quad \Rightarrow \quad e(5) = 0.225\,\text{mm}, \tag{3}$$

which corresponds to a ⌀0.45 zone at the top of the bracket, more than double the value the allocation in Table 2 assumed. Specifying ⌖ | ⌀0.2 Ⓟ5 | A | BⓂ moves the ⌀0.2 zone to the 5 mm the bracket actually occupies, so the allocation holds through the full mating thickness. It is inspected with a threaded gauge pin that protrudes 5 mm.

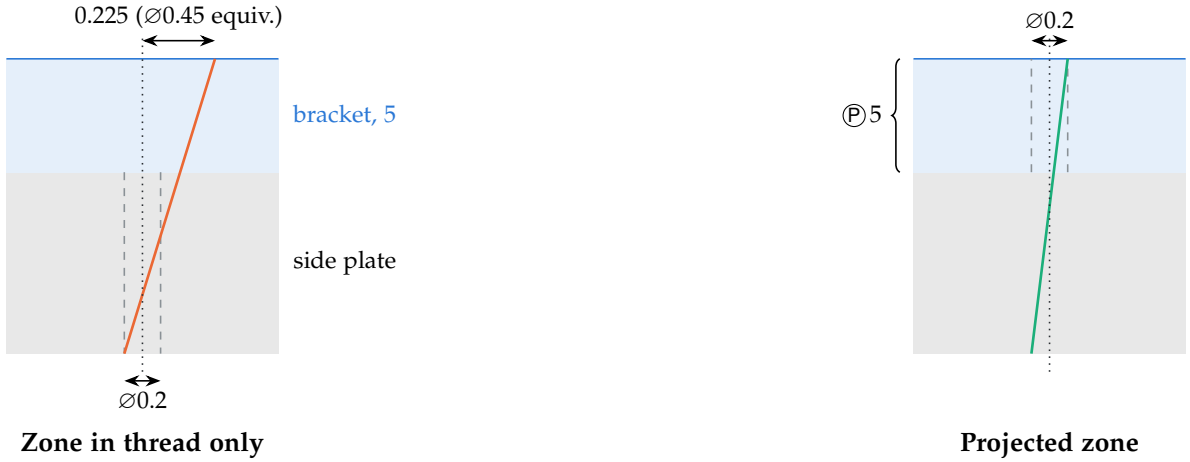


Figure 1: Worst-case thread axis. Left: a ⌀0.2 zone applied inside the 8 mm thread lets the screw axis drift to 0.225 mm from true position at the top of the bracket. Right: the projected zone holds ⌀0.2 through the bracket thickness.

6 Bracket coplanarity stack-up

6.1 Requirement

Each bracket is a flat plate bolted to two different parts: the side edge of this plate and the side face of the Static Connector. If those two faces are not coplanar, the bracket touches one part first and the screws must bend it into contact. That adds stress the FEA did not include and reduces the stiffness the brackets contribute; the FEA model assumed the faces were perfectly flush. The quantity of interest is therefore the offset g, normal to the bracket, between the side-plate edge and the Static Connector face.

6.2 Contributors

The two parts are located relative to each other only by the 4×M4 screws through the lower ∅4.4 holes, so the chain is: side-plate edge → side-plate holes → screws → Static Connector threads → Static Connector face.

Table 4: One-dimensional stack-up of the offset g (mm), holes at MMC.

#	Contributor	Source	±
1	Radial clearance of M4 in ∅4.4 hole	$(4.4 - 4.0)/2$	0.20
2	Side-plate 4× hole position	∅0.2 / 2	0.10
3	Static Connector thread position[a]	∅0.2 / 2	0.10
4	Side-plate edge profile	0.2 / 2	0.10
	Worst case $\sum t_i$		**0.50**
	RSS $\sqrt{\sum t_i^2}$		**0.26**

[a] Assumes the Static Connector side face is its datum, so its threads are located directly from it and the face contributes no separate term.

Two remarks qualify these numbers. First, the clearance term is not random in the statistical sense: it is set by how the assembler pushes the parts before tightening, so in practice it behaves more like a bias than a normal variable, and the RSS value is optimistic unless assembly is controlled. Second, with the holes at LMC the extra clearance and the bonus tolerance each add 0.075 mm, raising the worst case to ±0.65 mm.

6.3 Mitigation

The clearance term is the largest, and it is also the only one that can be removed at assembly without changing any drawing:

- **Straightedge alignment (adopted for the prototype).** Clamp a precision straightedge across the side-plate edge and the Static Connector face, then torque the four M4 screws. This eliminates term 1: worst case ±0.30, RSS $\sqrt{3} \times 0.1 = \pm 0.17$. Cost: none.
- **Dowel pins.** Two reamed ∅3 dowels between the parts replace the screw clearance with a press or slip fit, making the alignment repeatable after disassembly. Worth adopting if the joint is serviced often.

- **Shim or face after assembly.** Shim the bracket on the low side, or skim both faces flush in one setup. Effective but adds a process step.

Slotting the bracket holes would ease in-plane misalignment between the four bracket screws, but it does not reduce g, which is normal to the bracket.

7 Inspection plan

- **Datum A flatness:** surface plate and dial indicator, or CMM, 0.1 mm.
- **6× and 4× ∅4.4 holes:** hole size with pin gauges (4.40 go / 4.55 no-go); position with a functional gauge: six ∅4.20 pins (datum B) plus four ∅4.20 pins at the basic locations, with the part resting on a flat datum A plate. A part that drops onto all ten pins is acceptable.
- **Side threads:** M4-6H go/no-go plug gauge; position with threaded gauge pins protruding 5 mm, measured on a CMM against the datum A/B frame.
- **Outer profile:** CMM scan of the side edges, lower edge and R35 arc, best-fit to the datum frame allowing the B Ⓜ shift; every point within ±0.1 mm.

8 Conclusions

1. The datum frame follows function: A is the seating face, B is the bolt pattern that locates the plate on the rotor, referenced at MMC so inspection permits the same shift the screws permit.
2. All position tolerances follow from $T_1 + T_2 = H - F$. The two ∅4.4 patterns split a 0.4 mm budget evenly at ∅0.2Ⓜ.
3. The existing ∅4.3 bracket holes leave the bracket only ∅0.1; opening them to ∅4.5 gives ∅0.3 with a few minutes of rework.
4. A projected zone of 5 mm on the side threads is required: without it the screw axis can drift to the equivalent of ∅0.45 at the top of the bracket.
5. The bracket faces can be offset by up to ±0.50 mm (RSS ±0.26 mm). Straightedge alignment before torquing removes the dominant clearance term and reduces this to ±0.30 mm (RSS ±0.17 mm).

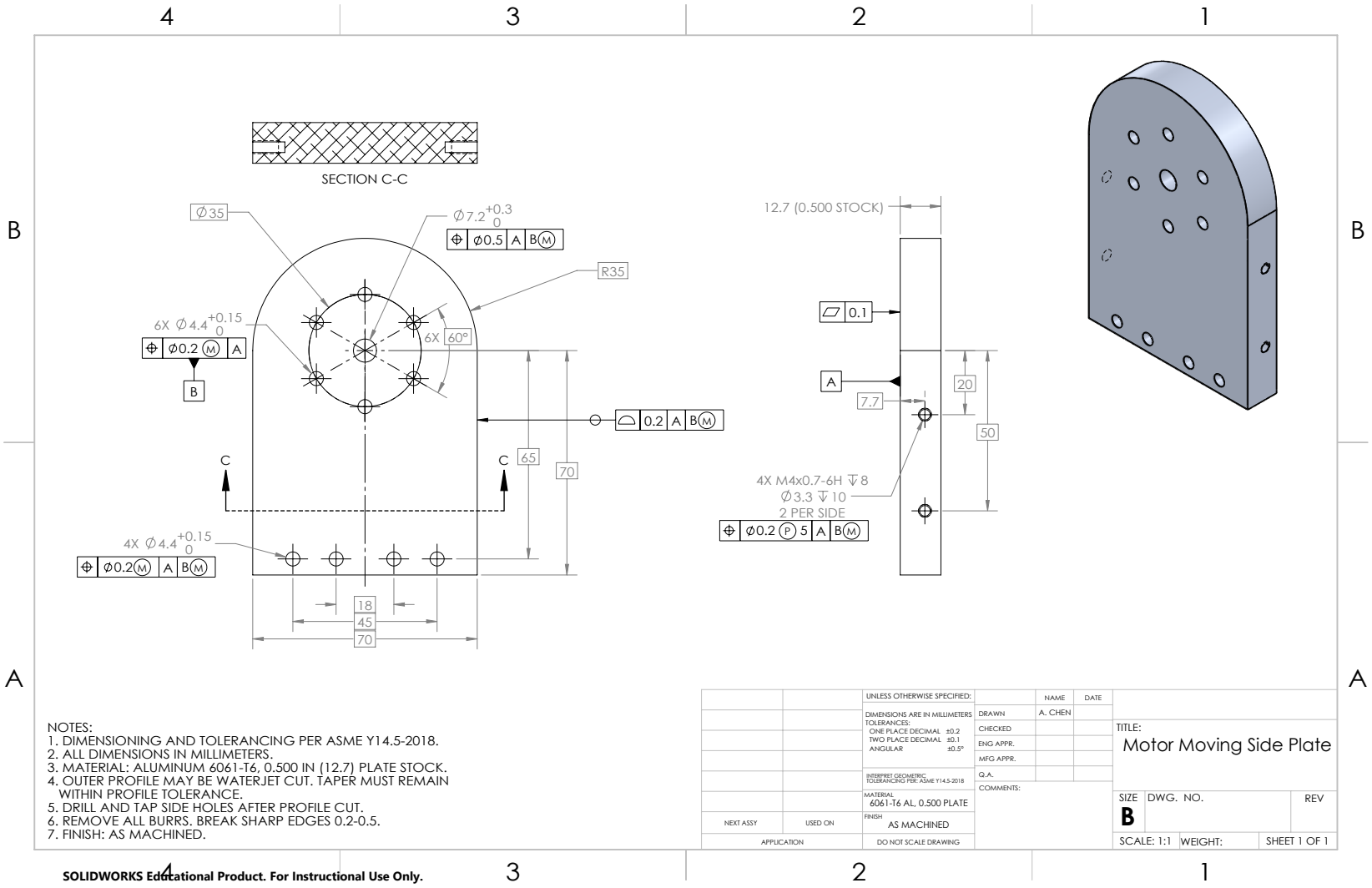


Figure 2: ASME Y14.5-2018 drawing of the Motor Moving Side Plate (B size, scaled to fit).